SIMPLE INTEREST & COMPOUND INTEREST

1. $SI = \frac{PTR}{100}$

P → principal → original money.

T → Time in years

R → Rate of interest.

2. $A = P + SI$ A → Amount to be paid.

1. Raju took a loan from a lendor at 12% per annum SI. After 3yrs, he had to pay Rs 5400 as interest. Find the principle borrowed by him.

$\boxed{\text{Ans: } 15000}$

$SI = \frac{PTR}{100}$ $P = \frac{SI \times 100}{TR}$

$5400 = P \times 3 \times \frac{12}{100}$ $P = \frac{\cancel{5400} \times 100}{\cancel{3} \times \cancel{12}} = 15000$

(or)

P → 100% always SI is always calculated on P

⇓

12% for 3years = 36% SI is same for all years

36% = 5400

100% = x

36x = 5400×100

$x = \frac{5400 \times 100}{36}$ $\boxed{x = 15000}$

2. A Farmer borrowed Rs 3600 at 15% Per annum SI. At the end of four years he cleared the debt by paying Rs. 4000 and a cow. Find the cost of the cow.

$\boxed{\text{Ans: } 1760}$

$100\% \rightarrow 3600$

$SI = 15\% \times 4 = 60\%$

$160\% \rightarrow x$

$A = P + SI = 100\% + 60\% = 160\%$

$$\frac{3600}{100} \times 160 = \underbrace{5760}_{\text{total amount to be paid}}$$

$5760 = 4000 + \text{cow}$

$\boxed{\text{cow} = 1760}$

3 A person borrowed some amount from a bank. Due to the fall in the rate of interest from 8% to $7\frac{3}{4}\%$. he has to pay Rs. 61.50 less. Find the capital he borrowed from the bank.

$\boxed{\text{Ans: } 24600}$

$$\frac{P \times 1 \times 8}{100} - \frac{P \times 1 \times 31}{4 \times 100} = 61.50$$

$\frac{1}{4}\% \rightarrow 61.50$

$1\% \rightarrow 61.50 \times 4 = 246$

$100\% \rightarrow 246 \times 100 = 24600/-$

4. At the rate of $8\frac{1}{2}\%$ per annum SI. The sum of rupees 4800 will earns how much Interest in 2 years 3 months

$\boxed{\text{ANS: } 918/-}$

2 yrs 3 months = $2 + \frac{3}{12} = \frac{9}{4}$ yrs.

$8\frac{1}{2}\% = 17/2\%$.

$$SI = \frac{\cancel{4800} \times 9 \times 17}{\cancel{4} \times \cancel{2} \times 100} = 918/-$$

5. Find the SI on Rs. 1820 from march 9th 2012 to May 21 2012. At $7\frac{1}{2}\%$ per annum SI. Find

$\boxed{\text{Ans: } 27.30/-}$

March 9 2012 to May 21 2012

March - 23

April - 30

May - 20

73 day

$$T = \frac{73}{365} = \frac{1}{5} \text{ years.}$$

$$SI = \frac{1820 \times 1 \times 15}{5 \times 2 \times 100} = \frac{273}{10} = 27.30/-$$

6. A person borrows Rs. 5000 for 2 yrs at 4% per annum SI. He immediately lends it to another person at $6\frac{1}{4}\%$ per annum for 2 yrs. Find his gain in the total transaction

$\boxed{\text{Ans: } 225/-}$

100% → 5000

1% → 50

$9/4\% \to \frac{9}{4} \times 50 \times 2 = 225/-$

gain = $6\frac{1}{4} - 4\%$

$= 2\frac{1}{4}\% = \frac{9}{4}\%$ percent.

7. If A, B, C are sums of money such that B is simple interest on A and C is simple interest on B. for same time and same rate of interest. Find the relationship between A, B and C.

$\boxed{\text{Ans:- } b^2 = ac}$

$\boxed{SI \propto P}$

$$\frac{SI_1}{SI_2} \propto \frac{P_1}{P_2} \Rightarrow \frac{b}{c} = \frac{a}{b}$$

$$\boxed{b^2 = ac}$$

8. Vishal invested Rs. 1000 at 5% PA (per annum) SI. If the interest is added to the principle after every 10 years. In how many years, the amount will become Rs. 2000.

$\boxed{\text{Ans: } 16.66 \text{ years}}$ (or) $16\frac{2}{3}$

$\boxed{P' = P + SI}$

10 years Rs 2000/-

↓

$P' = P + SI$

$$\boxed{R = \frac{SI}{P} \times 100\%}$$

Rate in 10 yrs,

$10 \times 5\% = 50\% = 500$

1yr ⟵ 5%

? ⟵ $\frac{100}{3}\%$

$P = 1500 \xrightarrow{+500} 2000$

From the formula,

$\frac{1}{5} \times \frac{100}{3} = \frac{20}{3} = 6\frac{2}{3}$

$$R = \frac{500}{1500} \times 100 = \frac{100}{3}\%$$

$T = 10 + 6\frac{2}{3}$

$T = 16\frac{2}{3}$.

9. A person lends 40% of his money at 15% per annum SI, 50% of the remaining at 10% PA SI. And Rest of the money at 18% PA SI. What is the effective annual rate of interest on the whole sum?

$\boxed{\text{Ans: } 14.40\%}$

$$\frac{40 \times 1 \times 15}{100} + \frac{30 \times 1 \times 10}{100} + \frac{30 \times 1 \times 18}{100} = \frac{100 \times 1 \times R}{100}$$

$$600 + 300 + 540 = 100R$$

$$1440 = 100R$$

$$R = \frac{1440}{100}$$

$$\boxed{R = 14.40\%}$$

~~10.~~

100%

40% of 15%	30% of 10%	30% of 18%
$\div 10 \downarrow$ $\downarrow \div 10$	$\div 10 \downarrow$ $\div 10 \downarrow$	$\downarrow \div 10$ $\downarrow \div 10$
$\boxed{4 \times 1.5}$	$\boxed{3 \times 1}$	$\boxed{3 \times 1.8}$

$$\begin{array}{r} 40\% \\ 15\% \\ \hline 6\% \end{array} + \begin{array}{r} 30\% \\ 10\% \\ \hline 3\% \end{array} + \begin{array}{r} 30\% \\ 18\% \\ \hline 5.4\% \end{array} = 14.4\%$$

10. A sum was invested on SI at certain rate of interest for 2 years. Had it been put at 3% higher rate, it would have fetched Rs. 72 more. Find the principle.

$\boxed{\text{Ans: } 1200}$

$R\% \rightarrow SI$

$(R+3)\% \rightarrow SI + 72$

$3\% \times 2 = 6\%$

$6\% \longrightarrow 72$

$1\% \longrightarrow 12$

$P = 100\% \longrightarrow 1200$

(or)

$$72 = \frac{P \times 2 \times (R+3)}{100} - \frac{P \times 2 \times R}{100}$$

$$72 = \frac{P \times 2 \times R}{100} + \frac{P \times 2 \times 3}{100} - \frac{P \times 2 \times R}{100}$$

$$72 = \frac{6P}{100}$$

$$\boxed{P = 1200/-}$$

11. A sum of Rs. 2400 amounts to Rs. 3264 in four yrs at certain rate of interest of SI. If the rate of interest is increased by 1%, find the amount on the same sum in the same time.

$\boxed{\text{Ans: } 3360}$

P		A
2400	$\xrightarrow{R\%}$	3264
2400	$\xrightarrow{(R+1)\%}$	?

$\underbrace{R\%}_{4\text{ yrs}} + \underbrace{1\%}$. for 4 yrs $1\% \Rightarrow 4\%$

$3264 + 96$

4% 2400 is 96

$\boxed{= 3360}$

12. Rajesh took an loan of Rs. 20000 from a bank for 3 yrs with rate of interests 5%, 7%, 9% for 3 consecutive yrs respectively. What is the total amount he need to pay to the bank at the end of 3 yrs.

$\boxed{\text{Ans:- } 24200}$

$SI = SI_1 + SI_2 + SI_3 = \frac{20000 \times 5 \times 1}{100} + \frac{20000 \times 7 \times 1}{100} + \frac{20000 \times 9 \times 1}{100}$

$= \frac{10,0000 + 14,0000 + 18,0000}{100}$

$A = P + SI$

$= 20000 + \_\_\_$

a

$SI = 5 + 7 + 9 = 21\%$

$A = 100\% + 21\% = 121\%$

$100\% \rightarrow 20000$

$1\% \rightarrow 200$

$121\% \rightarrow 121 \times 200$

$\boxed{24200}$

13. A sum of money at SI amounts to Rs. 815 in 3yrs and to Rs 854 in 4yrs. Find the principle.

$\boxed{\text{Ans: } 698}$

Rs. 815 3yrs Rs 854 in 4yrs

$SI = 1yr = 39/-$ $[854 - 815]$

$SI \rightarrow 3yrs = 3 \times 39 = 117/-$

$A = 815$

$P + SI = 815$

$P = 815 - 117$ $\boxed{P = 698}$

14. A sum of money amounts to Rs 5200 in 5 years, and to Rs 5680 in 7yrs. at SI. Find the rate of interest per annum.

Ans: 6%

Rs. 5200 (A) 5yrs   Rs. 5680 yrs.

P = 4000

2yrs $\xrightarrow{SI}$ 480

5yrs $\rightarrow$ $\frac{480}{2} \times 5$

SI = 1200/-

$SI = \frac{PTR}{100}$

$1200 = \frac{4000 \times 5 \times R}{100}$

R = 6%.

15. A sum of money doubles itself in 12yrs. In how many years it will become 3 time of itself at SI?

Ans: 24 yrs

$A = SI + P \Rightarrow SI = P, \quad P + P, \quad SI = 2P$

3P

P   SI = 2P $\longrightarrow$ $\frac{SI_1}{SI_2} = \frac{T_1}{T_2} \Rightarrow \frac{P}{2P} = \frac{12}{T_2} \Rightarrow \boxed{T_2 = 24}$

16. If a sum of money at SI becomes 21 times of itself in 100 yrs then in how many years it will become 13 times of itself.

$\boxed{\text{Ans: 60 years}}$

$$\frac{S1}{S2} = \frac{T_1}{T_2} = \frac{20P}{12P} = \frac{100}{T_2} = T_2 \times 2\cancel{0}\cancel{P} = 10\cancel{0} \times 12\cancel{P}$$

$$T_2 = \frac{120}{2}$$

$$T_2 = 60$$

17. If the ratio of principle and amount in current yr at SI is 9 : 17. And after 22 years the ratio will become 15 : 43. Find the rate of interest.

$\boxed{\text{Ans. } 4\frac{4}{9}\%}$

P	SI	A
9	8	17
15	28	43

P	SI	
135	120	
135	252	) 22 yrs

$$SI = \frac{PTR}{100}$$

$$132 = \frac{135 \times 22 \times R}{100}$$

$$\boxed{R = 4\frac{4}{9}}$$

18. Ramesh invested Rs. 7500 at 11% per annum simple Interest. He further invested some amount at SI at 15% per annum. The total interest earned at the

and of 1yr is 12% per annum on both the amounts find the amount he invested at 15% per annum.

$\boxed{\text{Ans: } 2500}$

M-1

11% 15%

12%

3 : 1

-7500

3 : 1

⇓

$\boxed{2500}$

$SI_2 + SI_1 = SI$

11% pa + 15% pa = 12% pa

$$\frac{7500 \times 1 \times 11}{100} + \frac{x \times 1 \times 15}{100} = \frac{(7500 + x) \times 1 \times 12}{100}$$

$15x = 12x + (12 \times 7500) - (11 \times 7500)$

$3x = 7500$

$\boxed{x = 2500}$

19 A person invested some amount at 12% p.a SI. and another certain amount at 10% p.a SI. He received an yearly interest of Rs. 130. If he had interchanged the amounts invested, he would have got Rs. 4 more as interest. What is the amount he invested at 10% per annum SI?

$\boxed{\text{Ans: } 700/-}$

$12\% \, x + 10\% \, y = 130$

$12\% \, y + 10\% \, x = 134$

$22\% \text{ of } (x+y) = 264$

$\frac{22}{100}(x+y) = 264$

$x + y = 1200$

$\% R = \frac{SI}{P} \times 100\%$

$= \frac{130}{1200} \times 100$

$R = \frac{65}{6}\%$

12% 10%

$\frac{65}{6}$

$\frac{65}{6} - 10$ $12 - \frac{65}{6}$

$\frac{5}{6} : \frac{7}{6} \Rightarrow \boxed{5 : 7}$

$12P \longrightarrow 1200$

10% of pa $7P \longrightarrow 700/-$

20. A person divided a sum of Rs. 17200 into 3 parts & invested at the rate of 5%, 6% & 9% pa SI. At the end of 2yrs, he got the same interest on each part of the money. Find the amount he invested at 9% pa SI.

$$\boxed{\text{Ans: } 4000/-}$$

$$P \propto \frac{1}{R}$$

$$P1 : P2 : P3 = \frac{1}{R_1} : \frac{1}{R_2} : \frac{1}{R_3}$$

$$= \left[\frac{1}{5} : \frac{1}{6} : \frac{1}{9}\right] \times \underbrace{90}_{LCM.}$$

$$= 18 : 15 : 10 \rightarrow 43 \text{ parts}$$

43 parts $\longrightarrow$ 17200

$9\% \Rightarrow$ 10p $\longrightarrow$?

$$\Rightarrow \frac{17200}{43} \times 10.$$

$\Rightarrow$ 4000/-

COMPOUND INTEREST

SI 100000 10%

1^{st} yr = 10,000

2nd yr = 10,000

3rd yr = 10,000

30,000

1,30,000

CI 1,00,000 10%

1st yr = 10,000

2nd yr = 10,000 + 1000

= 11000

3rd yr = 10,000 + 1000 + 1100

= 12100

1,33,000

Amount to be paid:

$$A = P\left(1 + \frac{R}{100}\right)^n$$

$n \rightarrow$ No of times principle is compounded.

Compound Interest:

$CI = A - P$

$$= P\left(1 - \frac{R}{100}\right)^n - P$$

T = 2yrs compounded annually

n = 2

compounded half yearly

n = 4

compounded quaterly

n = 8

1. What is the CI on Rs. 48000 for 2yrs at 20% per annum if the interest is compounded annually.

$\boxed{\text{Ans: } 21,120/-}$

$CI = A - 48000$

$= 69120 - 48000$

$= 21,120/-$

$$A = P\left(1 + \frac{R}{100}\right)^2$$

$$= 48000\left(1 + \frac{1}{6}\right)^2$$

$= 69,120.$

(or) $100\% \rightarrow 48000$

$44\% \rightarrow 44 \times 480 \Rightarrow \boxed{21120}$

NOTE:

1. For 2 yrs or 2 times CI is calculated as

$\%\ CI = a + b + \frac{ab}{100}$. ∴ a, b are rate of interest for 1st & 2nd yr

2. For 3 times or 3 yrs CI is calculated as

$\%\ CI = a + b + c + \frac{abc}{10000} + \frac{ab + bc + ca}{100}$.

R%	2yrs / 2 times	3yrs	4yrs
5%	10.25%	15.7625%	-
10%	21%	33.1%	46.41%
20%	44%	72.8%	-

2. In 2yrs at SI the principle increases by 16%. What will be the CI earned on Rs. 25000 in 2yrs at same rate of interest?

$\boxed{\text{Ans: } 4160/-}$

2 yrs → 16%

1 yr → 8% = R

$CI = 8 + 8 + \frac{8 \times 8}{100}$

$= 16.64\%$

$100\% \rightarrow 25000$

$CI = 16.64\% \rightarrow 250 \times 16.64$

$= 4160/-$

3) CI on a certain sum at 1yr at 14% p.a compounded half yearly is Rs. 289.8. Find the simple interest at same rate of interest for 1y on same sum.

$\boxed{\text{Ans: } 280}$

$1yr \rightarrow 14\%$

$6mon \rightarrow 7\%$

$CI = 7 + 7 + \frac{7 \times 7}{100}$

$= 14.49$

$14.49 \longrightarrow 289.8$

$14\% \longrightarrow ?$

$\Rightarrow \frac{289.8 \times 14}{14.49}$

$\Rightarrow 280$

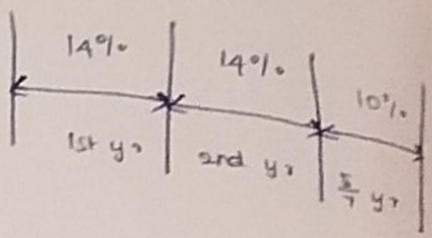
4. What is the CI on a sum of Rs 10000 at 14% p.a for $2\frac{5}{7}$ yrs if the interest is compounded annually.

$\boxed{\text{Ans: } 4295.6}$

100% ⟶ 10000

1% ⟶ 100

$2\frac{5}{7} \Rightarrow$



$CI = 14 + 14 + 10 + \frac{14 \times 14 \times 10}{100000} + \frac{140 + 196 + 140}{100}$

$14 \times \frac{5}{7} = 10.$

$CI = 38 + 0.196 + 4.76 = 42.956$

$CI = 42.926\% \longrightarrow 42.956 \times 100$ [1% → 100, 42.956% → 42.956 ×100].

$\longrightarrow \boxed{4295.6}$

5) What is the CI earned on Rs. 80000 at 40% pa for 1yr. if the interest is compounded quaterly?

$\boxed{\text{Ans: } 37128}$

1 yr ⟶ 40%

3mon ⟶ 10%

$CI = 80000(1 + 2/5)^4 - 80000$

$= 80000 \times \frac{24}{54} - 80000$

$CI = \underbrace{46.41\%}_{\text{in table.}}$

100% ⟶ 80000

46.41% ⟶ 46.41 × 800

⟶ 4641 × 8

⟶ 37128/-.

$\frac{2401}{625}$

6) Arun invested a certain amount at 8% p.a for 1st yr. and 9% p.a for 2nd yr. at CI. If ruppees 17,568 is received after 2 yrs, find the amount he invested.

$\boxed{\text{Ans: } 14923.541}$

$CI = 8+9+\frac{8\times9}{100}$

$= 17.72$

$17568 \rightarrow P + CI$

$100\% + 17.72\% \Rightarrow 117.72\%$.

$117.72\% \longrightarrow 17568$

$100\% \longrightarrow ?$

$\Rightarrow \frac{1756800}{117.72} \Rightarrow 14923.541\ldots$

7). If the difference b/w CI & SI on a sum of money for 3 yrs is Rs. 186. Find the principle if the rate of interest in both the cases is 10%.

$\boxed{\text{Ans: } 6000}$

$\left.\begin{array}{l}\%\ CI = 33.1\% \\ \%\ SI = 30\end{array}\right\}$ $3.1\% \longrightarrow 186$

$100\% \longrightarrow$

$\Rightarrow \frac{186}{3.1}\times 100$

$\Rightarrow 6000$

8) In how many months Rs 8000 gives Rs. 2648 as CI at 20 percent per annum compounded half yearly.

$\boxed{\text{Ans: 18 months}}$

$\% R = \frac{CI}{P} \times 100\%$

$= \frac{2648}{8000} \times 100$

$= 33.1\%$

12 mon $\rightarrow$ 20%

6 mon $\rightarrow$ 10%

$10\% \xrightarrow{\times 3 \text{ times}} 33.1\%$

6 mon $\times$ 3 time $=$ 18 months

9) A sum of money invested at CI amounts to Rs 800 in 3 yrs and to Rs. 882 in 5 yrs. What is the rate of interest per annum.

$\boxed{\text{Ans: 5\%}}$

3 yrs $\rightarrow$ 800

5 yrs $\rightarrow$ 882

$\Big\}$ 82 (2 yrs)

$\% R = \frac{82}{800} \times 100 = 10.25$ [for 2 yrs]

% R p.a $\rightarrow$ 5%.

(or)

$$\frac{P\left(1+\frac{R}{100}\right)^3}{P\left(1+\frac{R}{100}\right)^5} = \frac{800}{882} \Rightarrow \frac{1}{\left(1+\frac{R}{100}\right)^2} = \frac{400}{441} \Rightarrow \left(\frac{20}{21}\right)^2$$

$$\left(1+\frac{R}{100}\right)^2 = \left(\frac{21}{20}\right)^2 \Rightarrow 20R = 100 \Rightarrow \boxed{R = 5\%}$$

10) What is the rate of interest if SI earned on a certain amount for third year is Rs. 1750 & CI earned for 2yrs is Rs. 3622.50.

$\boxed{\text{Ans: } 7\%}$

SI

1st → 1750

2nd → 1750

CI

1st → 1750

2nd → 1750 + (3622.50 + 3500.00)

→ 1750 + 122.50.

$\%R = \frac{122.50}{1750} \times 100$

$\%R = 7\%$.

11). If the difference b/w CI and SI, on some amount at 20% p.a for 3 yrs is Rs. 48. Find the principle.

$\boxed{\text{Ans: } 375/-}$

CI and SI

↓ ↓

72.8% 3x20% = 60

(-)

12.8% = 48

$P = 100\% = \frac{48}{12.8} \times 100$

$\boxed{P = 375/-}$

12) What is the difference between CI on Rs. 12,500 for 1yr at 8% p.a rate of interest compounded halfyearly and yearly.

$\boxed{\text{Ans: } 20/-}$

half yearly   annually

4% @ 2times   8%

$4+4+\frac{16}{100}$

8.16%

12 mon → 8%

6 mon → 4%

T = 1yr.

100% → 12500

1% → 125

$0.16 \rightarrow \frac{12500}{100} \times 0.16$

⇒ 20 /-

13) A sum of money doubles itself at CI in 15 yrs. In how many years it will become 8 times of itself?

$\boxed{\text{Ans: } 45 \text{ yrs}}$

$P \xrightarrow{15yr} 2P \xrightarrow{15yr} 4P \xrightarrow{15yr} 8P$

×2   ×2   ×2.

⇒ 15 × 3 = $\boxed{45 \text{ yrs}}$

14. A sum of money placed at CI becomes 3 times of itself in 8 years. In how many years it will become 243 times of itself.

$\boxed{\text{Ans: } 40 \text{ yrs}}$

$P \xrightarrow{8yr} 3P \xrightarrow{8y} 9P \xrightarrow{8yr} 27P \xrightarrow{8yr} 81P \xrightarrow{8yr} 243P$

$\times 3 \quad \times 3 \quad \times 3 \quad \times 3 \quad \times 3$

$\Rightarrow 5 \times 8 = 40$ years

15) A certain sum of money becomes 2.25 times of itself in 2yrs. Find the rate of interest if its compounded annually.

$\boxed{\text{Ans: } 50}$

$P \longrightarrow \underset{\text{amount}}{2.25P}$

$P \qquad CI = 1.25P$

$\Rightarrow 50 + 50 + \frac{50 \times 50}{100}$

$\Rightarrow 100 + 25$

$= 125$. so $\Rightarrow \boxed{x = 50}$

$\% R = \frac{CI}{P} \times 100\%$

$= 1.25 \times 100\%$

$\% R = 125\%$.

16. A certain sum of money becomes $\frac{512}{162}$ times of itself in 4 years Find the rate of interest if it is compounded annually.

$\boxed{\text{Ans: } 33.33\%}$

$P \longrightarrow \frac{512}{162} = \frac{256}{81} = \frac{(4)^4}{(3)^4} = \left(\frac{4}{3}\right)^4$. in 4yrs

P

$P \xrightarrow{1yr} \frac{4}{3}P \xrightarrow{1yr} \left(\frac{4}{3}\right)^2 P \xrightarrow{1yr} \left(\frac{4}{3}\right)^3 P \xrightarrow{1yr} \left(\frac{4}{3}\right)^4 P$

$\times 4/3 \quad \times 4/3 \quad \times 4/3 \quad \times 4/3$

$$P \xrightarrow{1yr} \frac{4}{3}P \begin{cases} \nearrow P \\ \searrow \frac{1}{3}P = CI \end{cases}$$

$$\frac{CI}{P} = \frac{1}{3}$$

$$\% R = \frac{1}{3} \times 100 \boxed{= 33.33\%}$$

17. The amount received at 8% p.a CI after 2 yrs is Rs. 72900. Find the principle.

18. A sum of money becomes Rs. 64,800 at compound interest. If rate of interest for 3 yrs is 12.5 ~~yr~~ %, $6\frac{2}{3}\%$, 9.09% respectively. Find the compound interest.

17). $\boxed{\text{Ans: } 62500}$

$$\frac{8}{100} = \frac{4}{50} = \frac{2}{25} \begin{cases} \to CP \\ \to P \end{cases}$$

P : A

25 : 27

$25^2 : 27^2$

625 : 729.

62500 : 72900.

$\Rightarrow$ 62500

18). [The compounded ratio (or) combined ratio of a:b, c:d, e:f is ace : bdf.]

$\boxed{\text{Ans: } 15,300/-}$

12.5%

$\frac{1}{8} \begin{matrix} \to CI \\ \to P \end{matrix}$

$6\frac{2}{3}\%$

$\frac{20}{3}\%$

$\frac{20}{3} \times \frac{1}{100}$

$\frac{1}{15} \begin{matrix} \to CI \\ \to P \end{matrix}$

9.09%

$\frac{1}{11} \begin{matrix} \to CI \\ \to P \end{matrix}$

	P	:	A
$1^{st} \to$	8	:	9
$2^{nd} \to$	15	:	16
$3^{rd} \to$	11	:	12

$\frac{8 \times 15 \times 11}{9 \times 16 \times 2} = \frac{55}{72}$.

$\Rightarrow P : A = 55 : 72$.

$A = P + CI$

$CI = A - P$.

$72P \to 64800$

$17P \to ?$

$\Rightarrow \frac{64800}{72} \times 17$

$\Rightarrow$ 15,300/-